Wonjun Choi

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Fields

Econometrics Theory, Causal Inference, Machine Learning

Education

Emory University — **Atlanta, USA**
Ph.D. in Economics — *Spring 2027 (Expected)*

Korea University — **Seoul, Korea (South)**
Masters in Economics — *Spring 2021*
B.S. in Mathematics and Economics — *Spring 2018*

- Includes 2 years of military service.

Job Market Paper

A Machine Learning Approach to Affine Conditional Restriction Models

Conditional moment restriction (CMR) models provide a flexible framework for structural and reduced-form analysis. This paper develops a general framework for nonparametric estimation and inference in affine CMR models. We build on the sieve minimum distance framework of Chen and Pouzo (2015), while allowing modern machine learning methods in the first stage to accommodate flexible estimation of conditional moments with many instrumental variables. Using an orthogonalized minimum distance criterion, we characterize the requirements on first-stage learners through their prediction errors, without restricting the analysis to a particular class of learners. We provide a Wald statistic for testing smooth functionals and a Gaussian coupling of the estimated sieve coefficients, which can be used for various inference problems. We also provide lower-level verification of the first-stage requirements for selected machine learning methods.

Working In Progress

A Nonparametric Test of Supermodularity and Complementarity with Zhenting Sun and Zheng Fang

This paper develops a nonparametric test of supermodularity and complementarity.

Partially Constrained Deep Learning with Qihui Chen and Zheng Fang

This paper develops neural networks that enforce shape constraints by construction.

(Tentative) Distributional Difference in Differences with Ka Yan Cheng, Florian Gunsilius

We propose a distributional difference-in-differences framework that includes quantile difference-in-differences and other methods as a special case.

Research Experience

Department of Economics, Emory University — Atlanta, GA
Research Assistant to Prof. Stefan Hoderlein — Spring 2026
Research Assistant: Prof. Zheng Fang — Fall 2025
Research Assistant: Prof. Maria Arbatskaya — Spring 2024, 2025
Research Assistant: Prof. Caroline Fohlin — Fall 2023
Research Assistant: Prof. Ian McCarthy — Fall 2022

Department of Political Science, Korea University — Seoul, Korea (South)
Research Assistant to Prof. Woochang Kang — Fall 2020

Korea Labor Institute — Sejong, Korea (South)
Research Assistant to Ph.D. Enhye Kwak — Spring 2020

Department of Economics, Korea University Seoul, Korea (South)
Research Assistant to Prof. Jinyoung Kim Fall 2019

Teaching Experience

Department of Economics, Emory University Atlanta, GA
Teaching Assistant (ECON 524: Big Data Econometrics; Master) Fall 2025
Instructor (ECON 220 Lab: Data Science for Economists; UG) Spring 2025
Teaching Assistant (ECON 220: Data Science for Economists; UG) Fall 2024

Department of Economics, Korea University Seoul, Korea (South)
Teaching Assistant (Econometric Analysis; Graduate) Spring 2021
Teaching Assistant (Microeconometrics; Graduate) Spring 2020

Conference Presentations

Southern Economic Association (SEA) annual meeting *Huston* *November 2026*

Skills

Software Python, R, MATLAB LaTeX

Language English (fluent), Korean (native)

References

Please request reference letters from all writers through the proxy **econjm@emory.edu**

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Department of Economics
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Esfandiar Maasoumi
Department of Economics
Emory University

Pedro H.C. Sant'Anna
Department of Economics
Emory University